1 Revision of basic thermodynamics

Here are some questions which are designed to remind you about some basic classical thermal physics that you may have forgotten. In particular, they will give you some practice in working with the first law of thermodynamics and manipulating the equations $PV = n\mathrm{R}T$, $PV^\gamma = \text{constant}$ with $c_p/c_V = \gamma$, work done by gas is $\int P\mathrm{d}V$.

Useful data:

- Gas constant, $\mathrm{R} = 8.31$ J mol^{-1} K^{-1},
- Boltzmann's constant, $\mathrm{k_B} = 1.38 \times 10^{-23}$ J K^{-1},
- Molecular weight of ^{4}He = 4 and N_2 = 28 (these are in g mol^{-1}).
- 0°C $= 273$ K.
- STP (standard temperature and pressure) means $T = 0$°C and $P = 1$ atm (1 atm $\sim$ 1 bar $\simeq 10^5$ Pa).
- 1 litre = 10^{-3} m^3.

Problems indicated ♣ are a little harder.

Problem 1.1 **Ideal gas revision**
Numerical answers in square brackets.

(i) What volume does 1 kilomole of gas occupy at STP?

(ii) How many molecules does 1 mol of gas contain?

(iii) A 5 litre vessel contains 3 mol of gas at 200 K. What is the pressure? [997 000 Pa]

(iv) 3 mol of an ideal monoatomic gas is at a temperature of 60°C. What is

  (a) the internal energy, U? [12.5 kJ]
  (b) the heat capacity at constant volume, C_V? [37.4 J K^{-1}]
  (c) the heat capacity at constant pressure? [62.3 J K^{-1}]
  (d) why is C_p greater than C_V?

Problem 1.2 **Balloon**
A balloon contains 200 kg of helium gas. At launch the pressure is 1030 mbar and the temperature is 17°C. Assuming that the pressure in the balloon is the same as the surrounding air:

(i) What is the volume of the balloon? [1.17×10^3 m^3]

(ii) When the balloon has risen to its maximum height, where the pressure is 200 mbar and the temperature is only 5°C, what is the new volume? [5.78×10^3 m^3]

(iii) What does the assumption about the equality of pressures inside and outside imply about the material the balloon is made of?

Problem 1.3 **Isothermal and adiabatic processes**
0.5 m^3 of gas at 4 bar is compressed until the pressure is 8 bar. Given that $\gamma = 1.4$, what is the new volume if

(i) the process is isothermal? [0.25 m^3]

(ii) the process is isentropic? [0.30 m^3]

(iii) What does the value of $\gamma = 7/5$ imply if the gas is ideal?

Problem 1.4 **First law of thermodynamics revision – isothermal**
A quantity of gas at 120 bar, 0.8 m^3 and 25°C is compressed isothermally to 0.2 m^3. Calculate

(i) the change in internal energy of the gas,

(ii) the work done on the gas, [13.3 MJ]

(iii) the heat flow and which way it goes.

Problem 1.5 **First law of thermodynamics revision – adiabatic**
4 m^3 of gas at 50 bar and 120°C expands adiabatically to twice the initial volume. Molar heat capacities, $c_V = 20.8$ J K^{-1} mol^{-1}, $c_p = 29.1$ J K^{-1} mol^{-1}. Calculate

(i) the work done by the gas, by direct integration of the work, [12.1 MJ]

(ii) the final temperature, [25°C]

(iii) the change in energy, using part b). [12.1 MJ]

♣*Problem 1.6* **Frictional isothermal piston**
0.1 mol of an ideal gas in a cylindrical container of cross-sectional area 1 m^2 expands isothermally and quasi-statically at 20°C from 2 m^3 to 4 m^3, doing work against a piston. There is friction between the piston and the cylinder wall that provides a force of 10 N when the piston is moving. How much external heat is needed to accomplish the expansion assuming the heat generated by the friction all goes into the gas? [148.8 J]

What happens if one lowers the pressure towards zero?

♣*Problem 1.7* **A constant-force frictional piston**
An monoatomic ideal gas in a cylindrical container of cross-sectional area 1 m^2 expands quasi-statically from 2 m^3 to 4 m^3 while pushing on a piston that applies a constant force of 100 N on external machinery. How much external heat is needed to accomplish the expansion when:

(i) there is no friction between the piston and the cylinder wall?

(ii) the friction between the piston and the cylinder wall provides a force of 10 N when the piston is moving? Assume that the heat generated by the friction all goes into the gas. [530 J]